

Additional Experimental Result

March 2026

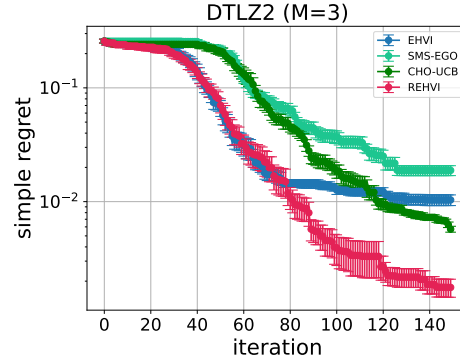


Figure 1: DTLZ2 with 6D input and 4 objectives: REHVI and CHO-UCB perform well relative to their base methods.

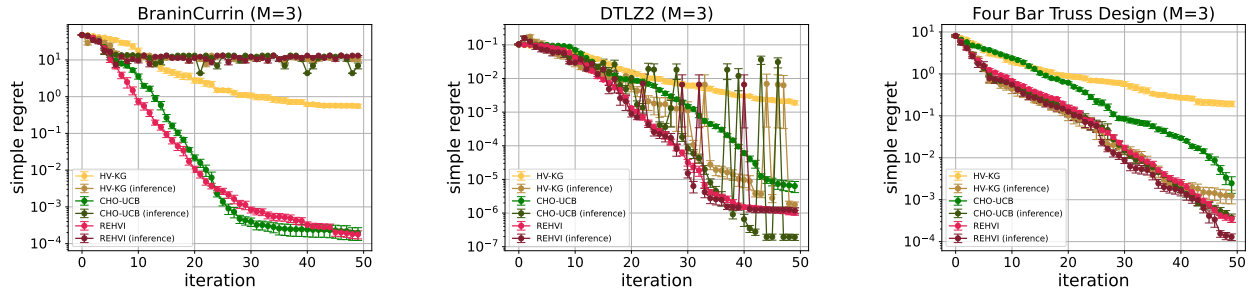


Figure 2: HVKG does not outperform the proposed methods in terms of observation regret. In terms of inference regret, all three acquisition functions perform similarly.

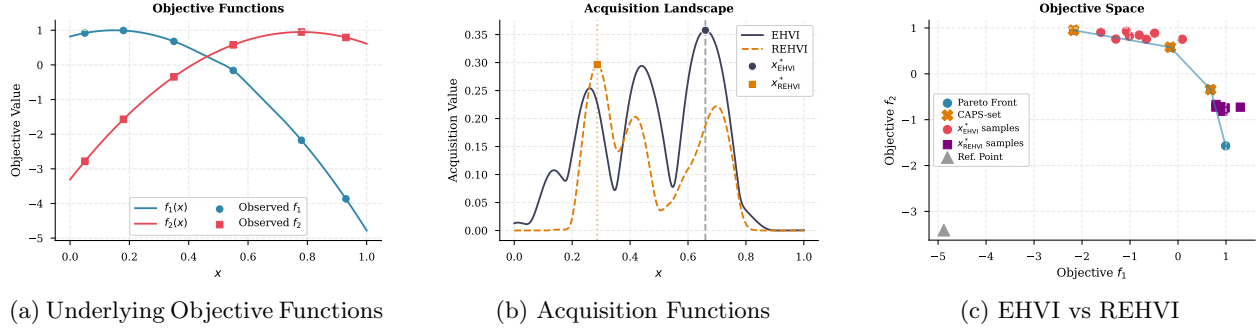


Figure 3: Illustration of EHVI and REHVI acquisition functions on a bi-objective toy problem: objective landscapes, acquisition values, and resulting candidate solutions in objective space.

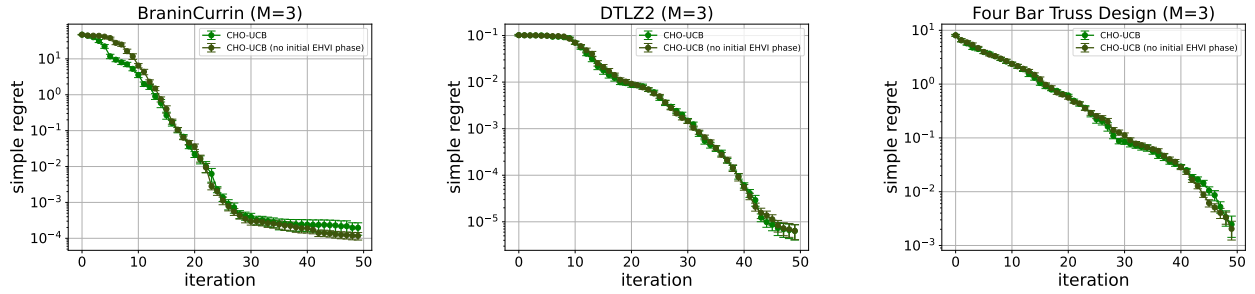


Figure 4: In Four-Bar Truss Design and DTLZ2, including or omitting the initial EHVI phase produces virtually identical results. In BraninCurrin, a minor early-stage divergence appears, but performance converges in the long run..